

Database System Engineering and Distributed Backend Development Course

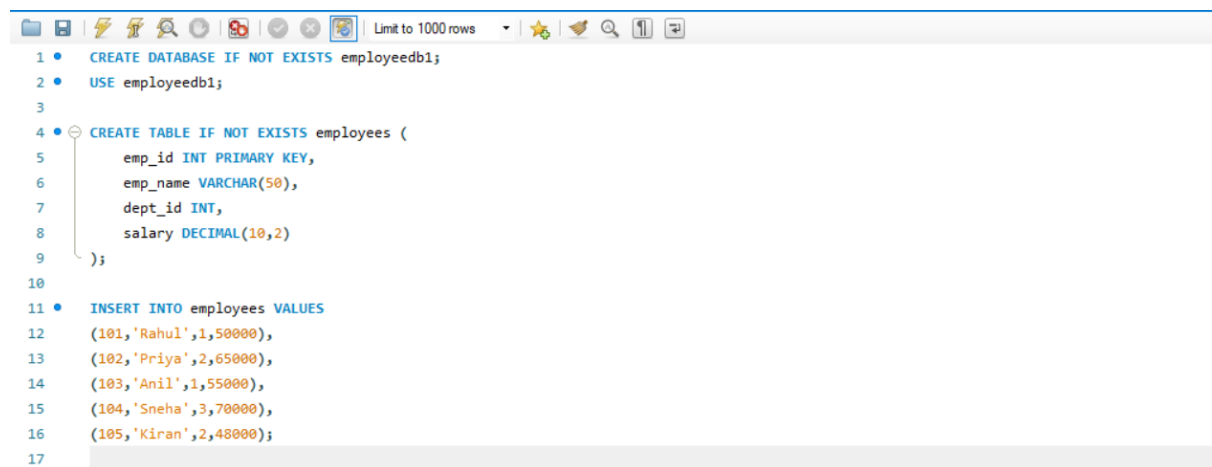
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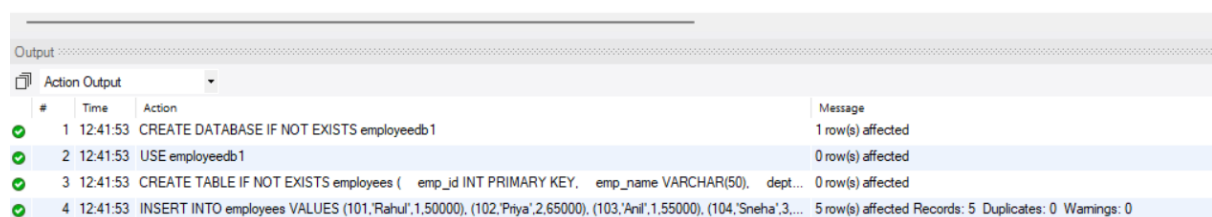
NAME : V SAI NIKHIL

WEEK-4 MYSQL STORED PROCEDURES

Setup: employee table



```
1 • CREATE DATABASE IF NOT EXISTS employeeeb1;
2 • USE employeeeb1;
3
4 • CREATE TABLE IF NOT EXISTS employees (
5     emp_id INT PRIMARY KEY,
6     emp_name VARCHAR(50),
7     dept_id INT,
8     salary DECIMAL(10,2)
9 );
10
11 • INSERT INTO employees VALUES
12     (101,'Rahul',1,50000),
13     (102,'Priya',2,65000),
14     (103,'Anil',1,55000),
15     (104,'Sneha',3,70000),
16     (105,'Kiran',2,48000);
17
```



#	Time	Action	Message
✓ 1	12:41:53	CREATE DATABASE IF NOT EXISTS employeeeb1	1 row(s) affected
✓ 2	12:41:53	USE employeeeb1	0 row(s) affected
✓ 3	12:41:53	CREATE TABLE IF NOT EXISTS employees (emp_id INT PRIMARY KEY, emp_name VARCHAR(50), dept...	0 row(s) affected
✓ 4	12:41:53	INSERT INTO employees VALUES (101,'Rahul',1,50000), (102,'Priya',2,65000), (103,'Anil',1,55000), (104,'Sneha',3,...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0

verify data

```
1 • SELECT * FROM employees;
```

Result Grid

Filter Rows:

Edit:

Ex

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	50000.00
	102	Priya	2	65000.00
	103	Anil	1	55000.00
	104	Sneha	3	70000.00
	105	Kiran	2	48000.00
✱	NULL	NULL	NULL	NULL

Display all employees

```
1 DROP PROCEDURE IF EXISTS GetEmployees;
2 DELIMITER $$
3
4 • CREATE PROCEDURE GetEmployees()
5 BEGIN
6     SELECT * FROM employees;
7 END $$
8
9 DELIMITER ;
10
11 • CALL GetEmployees();
12
```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	emp_id	emp_name	dept_id	salary			
▶	101	Rahul	1	50000.00			
	102	Priya	2	65000.00			
	103	Anil	1	55000.00			
	104	Sneha	3	70000.00			
	105	Kiran	2	48000.00			

in parametre

```
1 • DROP PROCEDURE IF EXISTS GetEmployeeById;
2 DELIMITER $$
3
4 • CREATE PROCEDURE GetEmployeeById(IN eid INT)
5 BEGIN
6     SELECT * FROM employees
7     WHERE emp_id = eid;
8 END $$
9
10 DELIMITER ;
11
12 • CALL GetEmployeeById(102);
13
14
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	emp_id	emp_name	dept_id	salary
►	102	Priya	2	65000.00

update salary

```
4
5 • CREATE PROCEDURE IncreaseSalary(
6     IN eid INT,
7     IN amount DECIMAL(10,2)
8 )
9 BEGIN
10     UPDATE employees
11     SET salary = salary + amount
12     WHERE emp_id = eid;
13 END $$
14
15 DELIMITER ;
16
17 • CALL IncreaseSalary(101,5000);
18 • SELECT * FROM employees WHERE emp_id=101;
19
```

Result Grid

	emp_id	emp_name	dept_id	salary
▶	101	Rahul	1	55000.00
*	NULL	NULL	NULL	NULL

Filter Rows: Edit:    Export

join

```

2 • DROP PROCEDURE IF EXISTS AvgSalary;
3 DELIMITER $$
4
5 • CREATE PROCEDURE AvgSalary()
6 BEGIN
7     SELECT AVG(salary) AS AverageSalary
8     FROM employees;
9 END $$
10
11 DELIMITER ;
12
13 • CALL AvgSalary();
14

```

Result Grid		Filter Rows:	Export:	Wrap Cell Content
	AverageSalary			
▶	58600.000000			

Group by and avg

```
8
9 • DROP PROCEDURE IF EXISTS EmployeeDepartment;
10 DELIMITER $$
11
12 • CREATE PROCEDURE EmployeeDepartment()
13 BEGIN
14     SELECT e.emp_name, d.dept_name, e.salary
15     FROM employees e
16     JOIN departments d ON e.dept_id=d.dept_id;
17 END $$
18
19 DELIMITER ;
20
21 • CALL EmployeeDepartment();
22
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

	emp_name	dept_name	salary
▶	Rahul	CSE	55000.00
	Priya	ECE	65000.00
	Anil	CSE	55000.00
	Sneha	EEE	70000.00
	Kiran	ECE	48000.00

If else :

```
4 • CREATE PROCEDURE DepartmentAverageSalary()  
5 BEGIN  
6     SELECT d.dept_name,  
7           COUNT(e.emp_id) AS EmployeeCount,  
8           AVG(e.salary) AS AverageSalary  
9     FROM employees e  
10    JOIN departments d ON e.dept_id=d.dept_id  
11   GROUP BY d.dept_id, d.dept_name;  
12 END $$  
13  
14 DELIMITER ;  
15  
16 • CALL DepartmentAverageSalary();  
17
```

Result Grid			
Filter Rows:		Export:	Wrap Cell Content:
	dept_name	EmployeeCount	AverageSalary
▶	CSE	2	55000.000000
	ECE	2	56500.000000
	EEE	1	70000.000000

```

8      SELECT salary INTO sal
9      FROM employees
10     WHERE emp_id=eid;
11
12     IF sal >= 60000 THEN
13         SELECT 'High Salary' AS Status;
14     ELSE
15         SELECT 'Low Salary' AS Status;
16     END IF;
17 END $$
18
19 DELIMITER ;
20
21 • CALL SalaryStatus(102);
22

```

Result Grid		Filter Rows:	Export:	Wrap Cell Co
	Status			
▶	High Salary			

Out parameter

Inout parameter

```
1 • DROP PROCEDURE IF EXISTS EmployeeCount;
2 DELIMITER $$
3
4 • CREATE PROCEDURE EmployeeCount(OUT total INT)
5 BEGIN
6     SELECT COUNT(*) INTO total
7     FROM employees;
8 END $$
9
10 DELIMITER ;
11
12 • CALL EmployeeCount(@total);
13 • SELECT @total AS TotalEmployees;
14
15
```

Result Grid		Filter Rows:	Export:	Wrap Cell Cont
	TotalEmployees			
▶	5			

transaction

```
1 • DROP PROCEDURE IF EXISTS AddBonus;
2   DELIMITER $$
3
4 • CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2))
5   BEGIN
6       SET bonus = bonus + 5000;
7   END $$
8
9   DELIMITER ;
10
11 • SET @bonus=10000;
12 • CALL AddBonus(@bonus);
13 • SELECT @bonus AS UpdatedBonus;
14
15
```

Result Grid   Filter Rows: Export:  Wrap Cell Content: 

	UpdatedBonus
▶	15000.00

```

13      SET salary=salary-amount
14      WHERE emp_id=fromEmp;
15
16      UPDATE employees
17      SET salary=salary+amount
18      WHERE emp_id=toEmp;
19
20      COMMIT;
21  END $$
22
23  DELIMITER ;
24
25 • CALL TransferSalaryAmount(101,102,2000);
26 • SELECT * FROM employees WHERE emp_id IN (101,102);
27

```

Result Grid	Filter Rows:	Edit:	Export/Import:
emp_id	emp_name	dept_id	salary
101	Rahul	1	53000.00
102	Priya	2	67000.00
NULL	NULL	NULL	NULL

```

1 • SHOW PROCEDURE STATUS
2 WHERE Db='employeeedb1';
3
4 SHOW CREATE PROCEDURE GetEmployees;
5
6 • DROP PROCEDURE IF EXISTS GetEmployees;
7

```

Procedure	sql_mode	Create Procedure	character_set_client	collation_connection	Database Collation
GetEmployees	ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLE...	CREATE DEFINER='root'@'localhost' PROCE...	utf8mb4	utf8mb4_0900_ai_ci	utf8mb4_0900_ai_ci

CREATE DATABASE IF NOT EXISTS employeeedb1;
USE employeeedb1;

CREATE TABLE IF NOT EXISTS employees (
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
dept_id INT,

```
    salary DECIMAL(10,2)  
);
```

```
INSERT INTO employees VALUES  
(101,'Rahul',1,50000),  
(102,'Priya',2,65000),  
(103,'Anil',1,55000),  
(104,'Sneha',3,70000),  
(105,'Kiran',2,48000);
```